7. Submission Requirements

Final Project Specification and Grading Rubric

Expected Outcome

Students will demonstrate the ability to combine AI tools, external APIs, application integrations, and conditional automation logic to create useful, end-to-end workflows while actively maintaining data quality, system reliability, and human oversight.

Part I: Project Overview & Architecture

1. STUDENT IDENTIFICATION

Your submission must clearly state your identity at the top of the report.

- [ ] Student Full Name
- [ ] GRID Number

2. PROBLEM STATEMENT & DIAGRAM

Clearly define the business or operational problem your automation solves. Include a visual flowchart mapping the data lifecycle from the initial trigger to the final output, identifying where AI is utilized.

- [ ] Concise Problem Statement paragraph.
- [ ] High-level Workflow Diagram (SVG, Draw.io, or equivalent).

[Insert Architecture Diagram Here]

3. TOOLS AND INTEGRATIONS

Provide a comprehensive list of the technical stack utilized to build the pipeline.

- [ ] Primary automation platform or programming language used.
- [ ] Specific AI Models accessed (e.g., Gemini Pro, Claude, GPT-4).
- [ ] External APIs and third-party applications integrated (e.g., Slack, Google Sheets, CRM

Part II: Implementation, Testing & Reflection

4. AI CONFIGURATION & PROMPTS

Document how you instructed the AI to process your data. The reliability of an automated system depends heavily on prompt engineering.

- [ ] Exact system prompts and user instructions used in the workflow.
- [ ] Sample inputs provided to the AI.
- [ ] Expected raw output from the AI model (e.g., categorized JSON).

5. TESTING & ERROR HANDLING

Robust workflows anticipate failure. Detail how your system handles unexpected states and prove its functionality through rigorous testing.

- [ ] **Test Cases & Results:** Provide a table documenting at least three test scenarios (Standard, Edge Case, Invalid Input) and their outcomes.
- [ ] **Error-Handling Approach:** Explain mechanisms for handling API timeouts, malformed data, or unexpected AI hallucinations.
- [ ] **Human Oversight:** Describe the routing conditions that escalate complex or critical issues to a human reviewer.

6. SECURITY CONSIDERATIONS & DEMONSTRATION

Prove the workflow is secure and operational.

- [ ] **Security:** Briefly explain how API keys were secured and how PII (Personally Identifiable Information) is protected or scrubbed during processing.
- [ ] **Demonstration:** Include annotated screenshots of the working pipeline, or provide a link to a video demonstration showing the workflow processing a live request.

7. FINAL CONCLUSION & LEARNING OUTCOMES

Conclude your report with a brief retrospective on the development process.

- [ ] Summarize the value generated by the completed workflow.
- [ ] Detail primary learning outcomes, specifically regarding the integration of deterministic code with probabilistic AI models.
- [ ] Identify one potential future improvement for the system.